Physics potential : **Top quark physics**

LHeC@1.2 TeV vs. LHμC@3.7 TeV

Hamzeh Khanpour

AGH University of Kraków

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Outline: LHeC@1.2 TeV vs. LHμC@3.7 TeV

- Single top-quark production at a lepton-hadron collider
- Top-quark pair production at a lepton-hadron collider
- Acceptance cross-sections for semi-leptonic decay channel

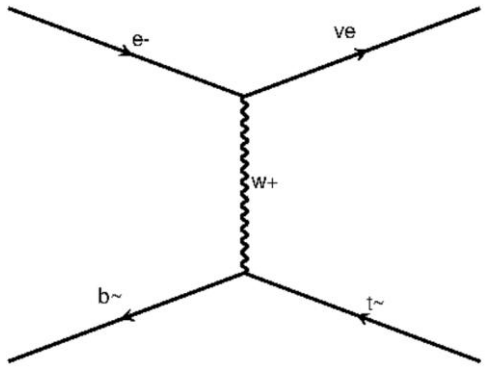
Single top-quark production at LHeC/LHμC

- Hard process (charged-current (CC) single-top) : At lepton-proton colliders, single-top production is dominated by CC DIS, mediated by a t-channel W exchange (5F scheme, b as a parton).

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p e- > t~ vl
```

$$e^- p \to \bar{t}\,\nu_e + X$$

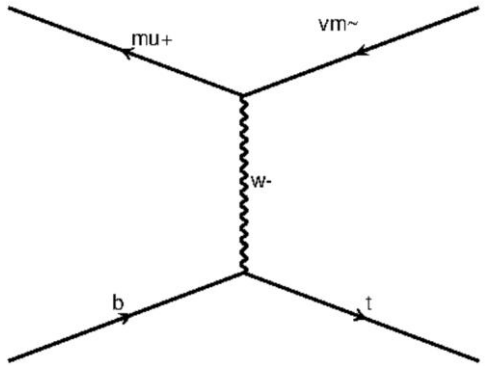
$$\sigma_{\mathrm{GEN}} = 1.6067304 \text{ pb}$$




```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p mu+ > t vl~
```

$$\mu^+ p \to t\,\bar{\nu}_\mu + X$$

$$\sigma_{\mathrm{GEN}} = 21.242199 \text{ pb}$$

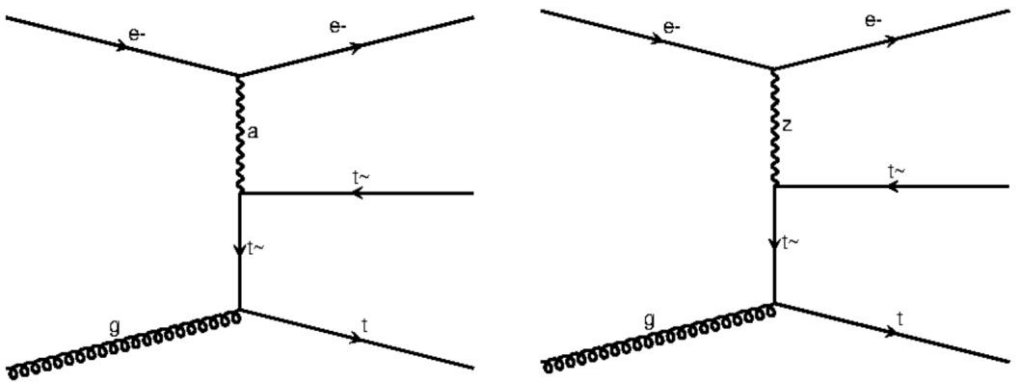


top-quark pair production at LHeC/LHμC

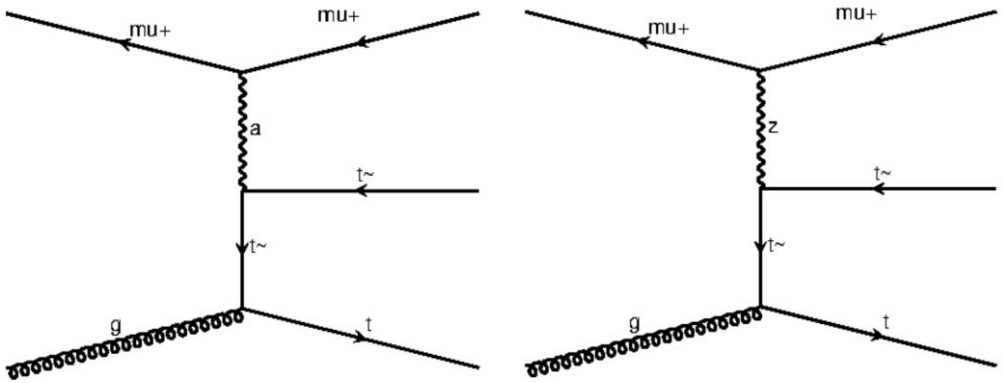
- At lepton-proton colliders, top-pair production is dominated by neutral-current (NC) interactions: the lepton emits a virtual photon (and also Z), which interacts with a gluon from the proton to produce the pair. [DIS-like: large Q^2 & Photoproduction-like: small Q^2]

- Underlying hard subprocess (dominant): $\gamma^* g \to t\bar{t}$ (and $Z^* g \to t\bar{t}$ is subleading)

$p\, e^- \to e^-\, t\bar{t}$



$p\, \mu^+ \to \mu^+\, t\bar{t}$



top-quark pair production at LHeC/LHμC

$$p\,e^- \to e^-\,t\bar{t}$$

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p e- > t t~ e-
```

$\sigma_{\mathrm{GEN}} = 0.018435$ pb

$$p\,\mu^+ \to \mu^+\,t\bar{t}$$

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p mu+ > t t~ mu+
```

$\sigma_{\mathrm{GEN}} = 0.96053$ pb

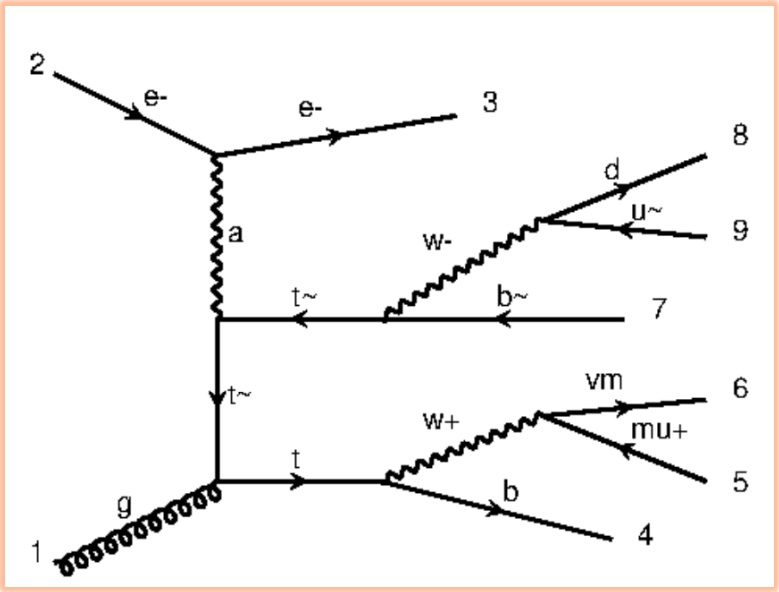
- GEN-level cuts actually applied (run card):

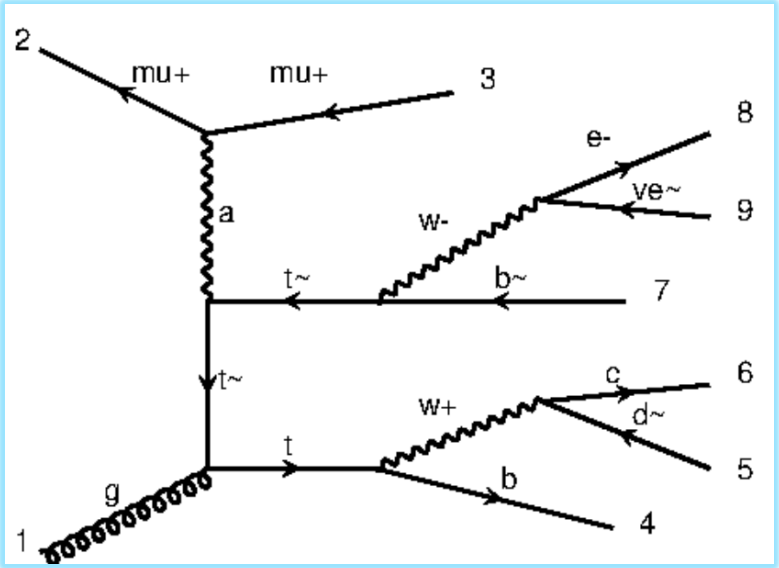
$$p_T^\ell > 0.1\ \mathrm{GeV}, \qquad |\eta_\ell| < 9$$

Acceptance cross-sections for semi-leptonic decay channel

- Physics goal: acceptance (fiducial) cross sections in the semi-leptonic channel

- The semi-leptonic mode gives a good balance:
- one charged lepton for clean tagging,
- hadronic side for full kinematic reconstruction.
- It is the standard channel to estimate realistic observable rates at GEN level before shower/detecto





- LHeC/LHmuC acceptance (Krzysztof requirement) [for the charged lepton and the four jets (2 b-jets + 2 light jets)]:

$$|\eta_\ell| < 5, \qquad |\eta_j| < 5$$

$$|\eta_\ell| < 4, \qquad |\eta_j| < 4$$

- Acceptance: $A = \frac{N_{\text{pass}}}{N_{\text{total}}}$

- Fiducial cross section: $\sigma_{\text{fid}} = \sigma_{\text{GEN}} \times A$

○ The semi-leptonic topology: One top decays leptonically and the other decays hadronically:

$$p\,e^- \to e^-\,t\bar{t}, \quad t \to bW^+,\ W^+ \to \ell^+\nu_\ell, \quad \bar{t} \to \bar{b}W^-,\ W^- \to jj$$

$$p\,e^- \to e^-\,t\bar{t}, \quad t \to bW^+,\ W^+ \to jj, \quad \bar{t} \to \bar{b}W^-,\ W^- \to \ell^-\bar{\nu}_\ell$$

- At matrix-element (truth) level we have:
- 1 scattered beam lepton (e^- or μ^+ (in the final state (NC process),
- 1 charged lepton from W decay,
- 1 neutrino from W decay,
- 4 quarks → 4 jets

Generator-level cuts

- Generator-level “loose” cuts
- Transverse momentum + pseudorapidity ranges + angular separations

$$p_T^j > 0.1, \qquad p_T^b > 0.1, \qquad p_T^\ell > 0.1 \text{ GeV}$$

$$|\eta_j| < 9, \qquad |\eta_b| < 9, \qquad |\eta_\ell| < 9$$

$$\Delta R_{jj} > 0.4, \qquad \Delta R_{j\ell} > 0.4, \qquad \Delta R_{\ell\ell} > 0.4$$

Acceptance cross-sections

LHeC

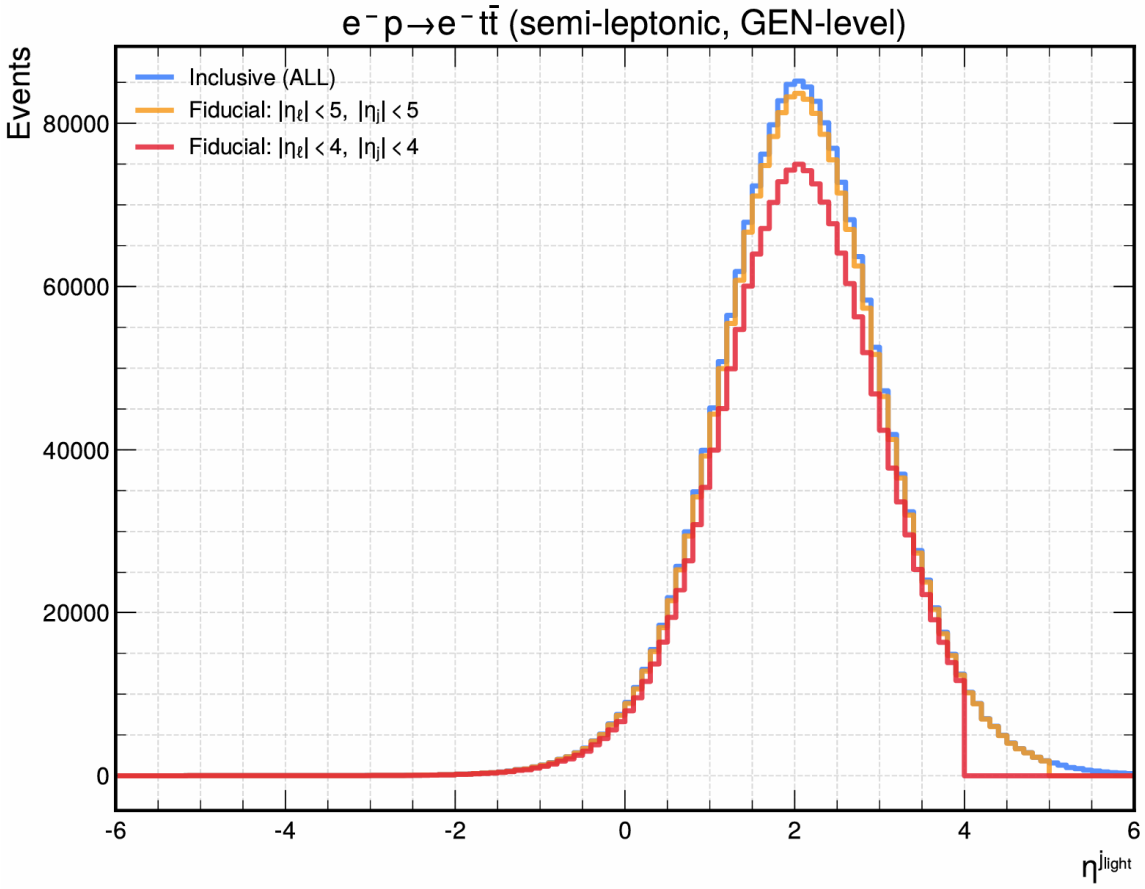
$$A(|\eta| < 5) = \frac{0.0887659}{0.0906378} \approx 97.93\%, \quad A(|\eta| < 4) = \frac{0.0780355}{0.0906378} \approx 86.10\%$$

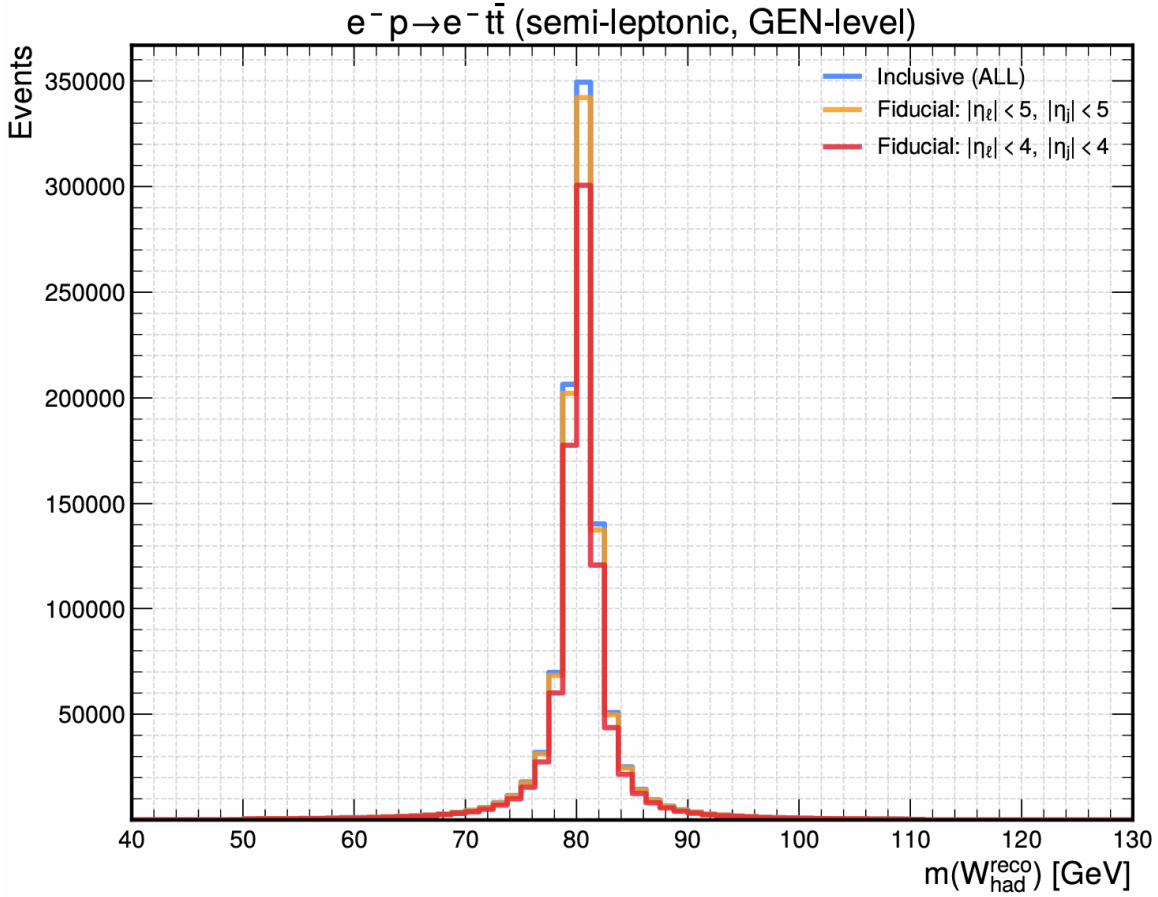
LHμC

$$A(|\eta| < 5) = \frac{7.81220}{7.83363} \approx 99.73\%, \quad A(|\eta| < 4) = \frac{7.68025}{7.83363} \approx 98.04\%$$

- Because this study is performed at generator level (no Pythia showering and no Delphes detector simulation), the resulting σ_{fid} isolates the geometric/kinematic acceptance for the semi-leptonic topology; more realistic efficiencies and migrations can be incorporated later by repeating the same procedure after showering, jet clustering, and detector reconstruction.

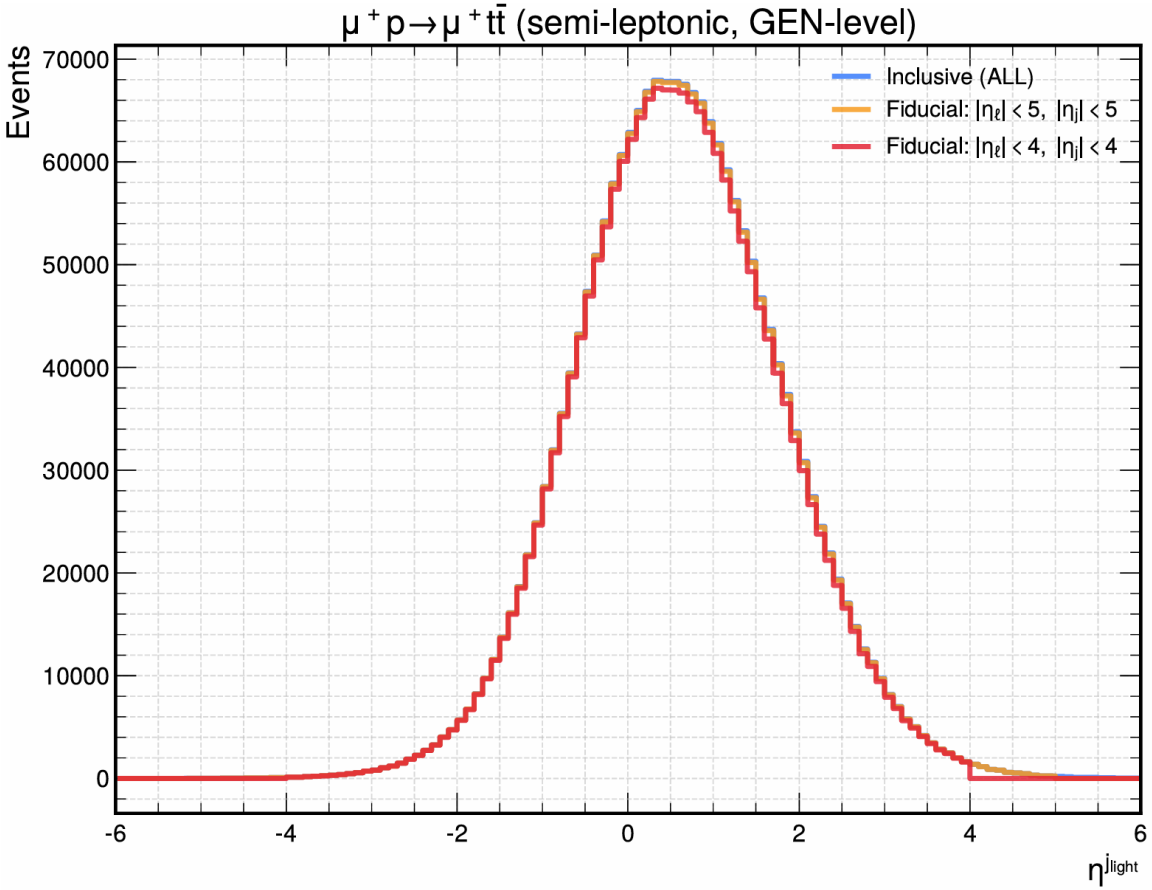
GEN-level acceptance at the LHeC

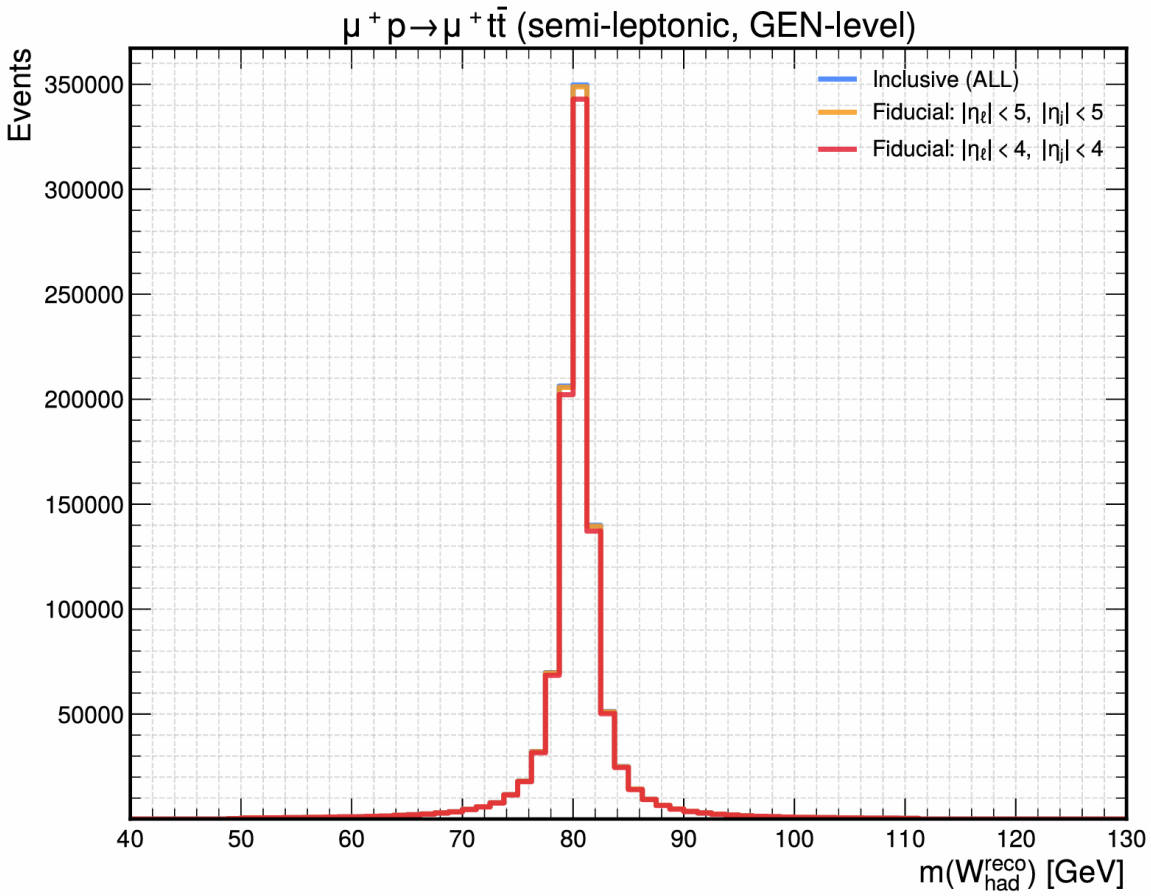




GEN-level distributions for the semi-leptonic $t\bar{t}$ channel in $e^{-}p \rightarrow e^{-}\ t\bar{t}$) :left) pseudorapidity of the light jet η_j^{light} , and (right) reconstructed hadronic W-boson mass $m\left(W_{had}^{reco}\right)$.

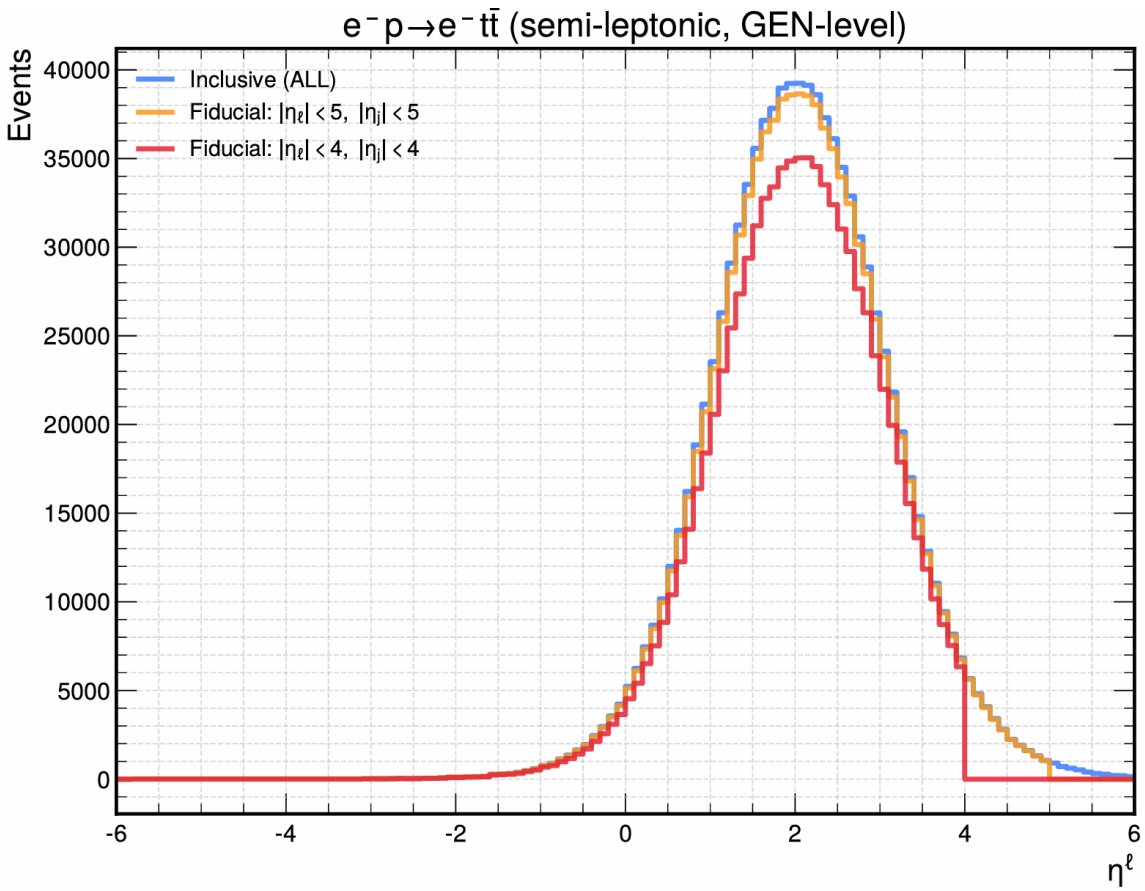
GEN-level acceptance at the LHμC

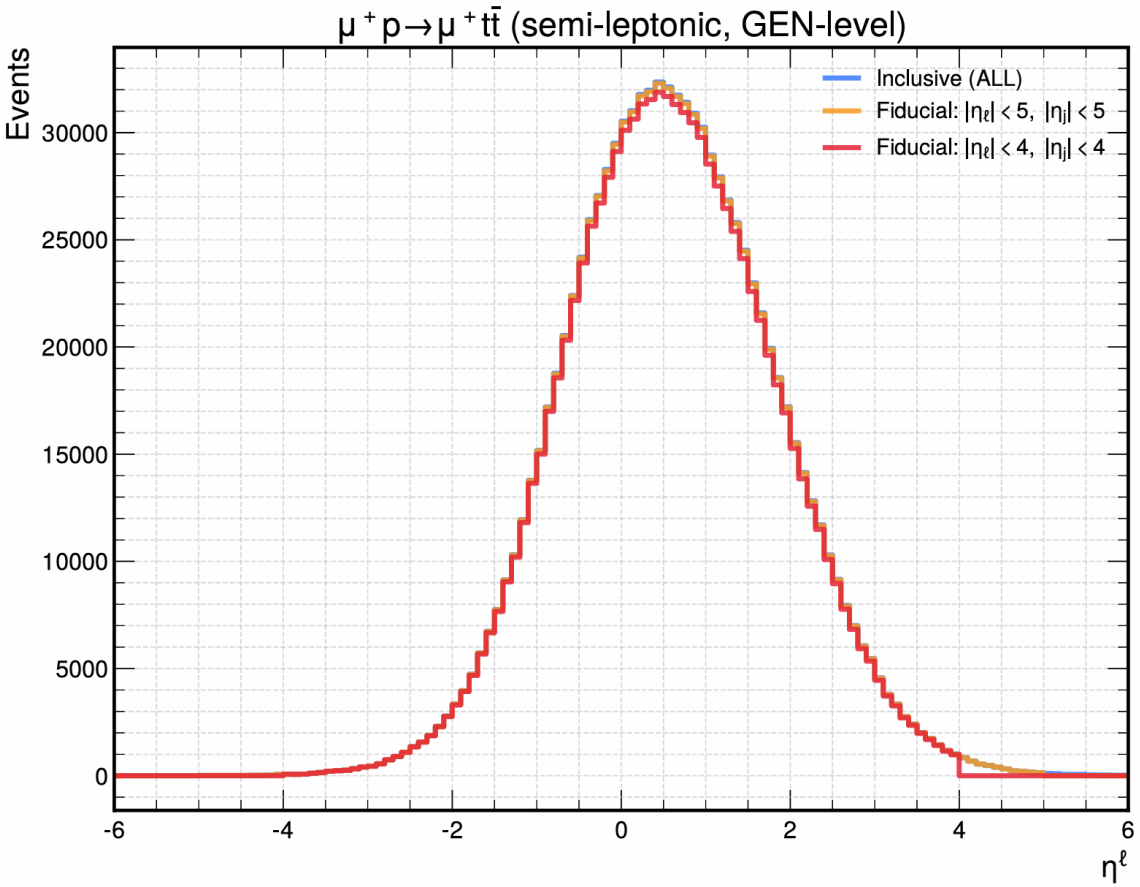




GEN-level distributions for the semi-leptonic $t\bar{t}$ channel in $\mu^+ p \to \mu^+ \, t\bar{t}$) :left) pseudorapidity of the light jet η_j^{light} , and (right) reconstructed hadronic W-boson mass $m\left(W_{had}^{reco}\right)$.

Lepton η acceptance in semi-leptonic $t\bar{t}$: LHeC vs LHμC





GEN-level distributions for the semi-leptonic $t\bar{t}$ channel in: (left) LHeC, and (right) LHμC.

LHμC gives the $t\bar{t}$ system more central production.

Thank
you!